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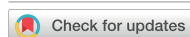
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Uncertainty in Antarctic precipitation projections under global warming

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E-mail: lixichen@mail.iap.ac.cn**Keywords:** Antarctic precipitation, uncertainty, future projection, CMIP6Supplementary material for this article is available [online](#)

Abstract

The Antarctic Ice Sheet is a critical driver of global sea level rise, yet future projections of Antarctic precipitation remain highly uncertain, posing challenges to modeling ice sheet changes. This study examines uncertainties in Antarctic precipitation projections using simulations from the Coupled Model Intercomparison Project Phase 6 and investigates their underlying sources. We find that uncertainties in precipitation projections are substantially larger than those for surface temperature. Integrated precipitation uncertainties are strongly linked to uncertainties in both global and Antarctic surface temperatures, while atmospheric circulation—particularly the Pacific South American modes—plays a critical role in regional precipitation patterns of precipitation uncertainties. Further analysis reveals that tropical sea surface temperatures contribute to uncertainties in Antarctic precipitation patterns through atmospheric teleconnections. These findings highlight the need to improve polar processes in climate models to reduce uncertainties in Antarctic precipitation projections and better predict ice sheet contributions to sea level rise.

1. Introduction

The Antarctic ice sheet has the potential to contribute substantially to future sea-level rise [1, 2] due to enhanced ice shelf melting, calving, and the subsequent acceleration of ice flow at its margins [3, 4]. However, these losses are expected to be partially mitigated by an increase in surface mass balance [5–7]. Given the cold climate of Antarctica, surface melt is largely confined to the ice sheet's peripheral regions [8, 9] and plays a minimal role in current and projected surface mass balance changes [10, 11]. Instead, anticipated increases in surface mass balance are primarily driven by enhanced precipitation across the continent [12–14]. Accurate future projections of integrated and regional precipitation are therefore critical not only for estimating ice accumulation but also for modeling changes in the ice sheet's dynamic structure [15], which could affect ice shelf stability and influence the long-term evolution of the Antarctic ice sheet [16, 17].

Despite its significance, projections of Antarctic precipitation remain highly uncertain at both regional and continental scales [12–14]. These uncertainties primarily arise from deficiencies in, and discrepancies among, global climate models [18–20]. Assessing these uncertainties and identifying their underlying causes are critical for improving model performance and, where possible, supporting informed adaptation and mitigation planning [21, 22]. Future changes in Antarctic precipitation are related to atmospheric warming driven by rising CO₂ concentrations [23, 24]. Warmer air has a greater capacity to hold moisture, which enhances moisture transport to the Antarctic continent [24]. However, the magnitude of atmospheric warming—and its subsequent impact on Antarctic precipitation—varies considerably across models, even under the same emissions scenario [14, 25], leading to substantial uncertainty in projections. In addition to the thermal effects of atmospheric warming, variations in atmospheric circulation play a key role in dynamically

modulating the Antarctic climate. The Pacific South American (PSA) modes (PSA1 and PSA2) [26, 27] are particularly influential, affecting Antarctic sea ice [28], surface temperature [29, 30], and precipitation [31, 32] through both moisture and thermal advection, as well as ocean-atmosphere-sea ice feedbacks [33, 34]. Uncertainty in projections of mid-to-high-latitude atmospheric circulations in the Southern Hemisphere further compounds the challenges of Antarctic precipitation projections.

Recent studies reveal that regional atmospheric circulation mediates the remote influence of tropical sea surface temperature (SST) variations on the Antarctic climate. For example, the El Niño-Southern Oscillation [35] and the Indian Ocean Dipole [36] have been shown to strongly influence precipitation in the Antarctic Peninsula and West Antarctica [31, 37–39], as well as surface temperature in the Antarctic Peninsula [40]. In the future, tropical oceans are projected to warm, with an El Niño-like pattern in the tropical Pacific [41, 42] and pronounced warming anticipated in the tropical Indian and Atlantic Oceans [43, 44]. These changes, however, are associated with substantial model uncertainty [45], which may contribute to the Antarctic precipitation projections through atmospheric teleconnections [46].

In this study, we evaluate the model uncertainty in both integrated and regional Antarctic precipitation using multi-model outputs from the Coupled Model Intercomparison Project Phase 6 (CMIP6). To better understand the sources of this uncertainty, we investigate the relationship between uncertainty in Antarctic precipitation and uncertainties in surface temperature and atmospheric circulation through regression analysis and maximized covariance analysis (MCA). Furthermore, we examine the contribution of tropical SST uncertainty to Antarctic precipitation changes, highlighting the critical linkages between tropical and high-latitude climate systems under global warming.

2. Data and methods

2.1. Data

We use outputs from 34 models in CMIP6 (table S1) under the high-emission scenario SSP5-8.5 to assess projected changes in Antarctic precipitation during the 21st century. The SSP5-8.5 scenario, often referred to as the ‘fossil-fueled development’ pathway, represents a global mean CO₂ concentration of about 1135 ppm and a projected global warming of 5 °C by the end of the century [47]. Projected precipitation changes are quantified as the difference between the multi-model mean for 2081–2100 under SSP5-8.5 and the mean for 1981–2014 from historical simulations. Surface temperature is used

both as a reference for the magnitude of model uncertainty in precipitation changes and to explore its linkage to Antarctic precipitation. The Antarctic and global mean temperature changes are calculated as area-weighted averages of surface temperature. The 500 hPa geopotential height is used to evaluate the atmospheric circulation changes in the Southern Hemisphere. The SST data are also used to evaluate the potential relationships between the model uncertainties in Antarctic precipitation and tropical oceans. Since the native spatial resolutions of the CMIP6 models vary, all data were regridded to a uniform 1° × 1° latitude–longitude grid using the second-order conservative remapping scheme implemented in the Climate Data Operators (CDO) [48]. Conservative remapping is widely used in climate model evaluation and multi-model ensemble studies because it preserves the integral quantities and avoids artificial discontinuities that may arise from simpler methods such as nearest-neighbor or bilinear interpolation [49].

2.2. Methods

2.2.1. Statistical analysis

To assess the robustness of projected changes across models, we compute the signal-to-noise ratio (SNR), a commonly used metric for evaluating multi-model agreement in climate projections [50]. The ‘signal’ is defined as the multi-model mean change between the future (2081–2100) and historical (1981–2014) periods. The ‘noise’ is defined as the inter-model standard deviation of the same change field. The SNR is then calculated as the ratio of the multi-model mean change to the model standard deviation at each grid point. Higher SNR values indicate greater robustness and inter-model agreement, whereas low SNR values suggest large uncertainty relative to the mean response.

Ordinary least squares regression is used to analyze the relationship between Antarctic precipitation and tropical SST. We apply Student’s *t*-test [51] to estimate the confidence level of the regression coefficients. The degrees of freedom are adjusted according to the autocorrelation in the climate data. Ordinary least squares regression is also used to remove the component of Antarctic precipitation changes that are related to the Antarctic mean surface temperature.

We employed MCA to identify the coupled variability between Antarctic precipitation and atmospheric circulation. MCA quantifies the dominant patterns of covariability between two fields by performing a singular value decomposition of their cross-covariance matrix [52]. The resulting leading modes represent spatial patterns that maximize shared covariance between precipitation and atmospheric circulation, together with their associated time series. This approach enables us to isolate the

primary coupled modes contributing to regional differences in Antarctic precipitation.

2.2.2. PSA patterns extraction

In this study, we follow previous literature in defining PSA1 and PSA2 as the two leading patterns of the Pacific–South American teleconnection in the Southern Hemisphere [26, 27]. PSA1 corresponds to the canonical PSA wave train extending from the central tropical Pacific toward the Amundsen–Bellingshausen Sea, whereas PSA2 represents a longitudinally shifted variant with centers of action located farther toward the South Pacific and the New Zealand sector. Together, these two modes characterize the dominant extratropical atmospheric response to tropical Pacific SST anomalies. For both ERA5 and CMIP6, PSA1 and PSA2 are computed using monthly 500 hPa geopotential height anomalies over the Southern Hemisphere for latitudes south of 20° S. Anomalies are obtained by removing the monthly climatology. We then perform empirical orthogonal function (EOF) analysis on the monthly anomalies for the historical period (1981–2014). PSA1 and PSA2 are identified as the second and third leading EOFs (EOF2 and EOF3). The corresponding principal components (PCs) provide the temporal variability of PSA1 and PSA2. To obtain the annual PSA patterns, we compute the annual mean PCs and regress the corresponding annual-mean 500 hPa geopotential height field onto each annual PC. The resulting regression maps are taken as the PSA1 and PSA2 patterns. For CMIP6, the final PSA patterns are obtained by averaging the regression maps across all models.

3. Results

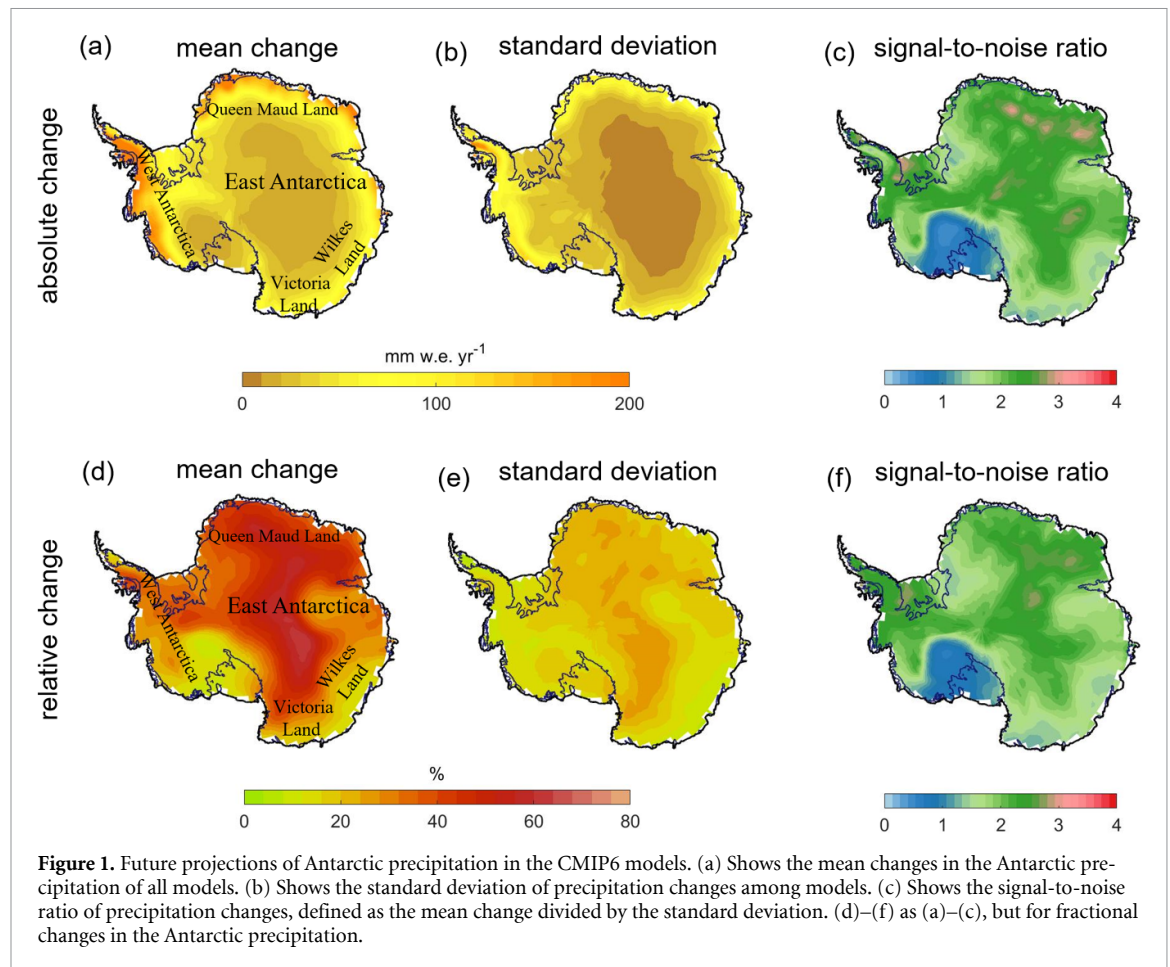
3.1. Future projections of Antarctic precipitation

Before assessing future projections, we first evaluated the performance of CMIP6 models in reproducing the present-day climatology of Antarctic precipitation and surface temperature by comparing the historical multi-model mean with ERA5 for the period 1981–2014 (figure S1). The CMIP6 multi-model mean (figures S1(b) and (e)) captures the major spatial features of Antarctic precipitation and temperature in ERA5 (figures S1(a) and (d)), indicating that the models reasonably reproduce the overall distribution of the Antarctic surface climate. However, notable biases exist in magnitude: the CMIP6 multi-model mean generally exhibits higher precipitation (figure S1(c)) and surface temperature (figure S1(f)) than ERA5. An examination of individual models shows substantial inter-model spread, with some models producing wetter conditions and certain others showing drier biases (figure S2). A cross-model comparison reveals a strong correlation ($r = 0.69$) between the biases in Antarctic mean precipitation and temperature (figure S3), suggesting that uncertainty in the simulated Antarctic precipitation is closely linked

to temperature biases across models. These results support the reliability of CMIP6 models for projecting future Antarctic climate changes, while highlighting systematic biases that need to be considered when interpreting future projections.

We evaluate the future projections of Antarctic precipitation changes by the end of the 21st century (2081–2100) under the SSP5-8.5 scenario using data from CMIP6 models, relative to present-day precipitation (1981–2014) (figures 1(a)–(c)). The mean projected increase in integrated Antarctic precipitation is 60 mm yr^{−1}, with a wide range of variability across models, from 24 mm yr^{−1} to 106 mm yr^{−1} (figure S4). These changes also exhibit notable regional disparities. Multi-model mean results indicate an overall increase in Antarctic precipitation, with the largest projected changes occurring in coastal regions, where precipitation is climatologically abundant, compared to smaller changes in the drier interior of East Antarctica (figure 1(a)). However, certain regions, such as the western part of West Antarctica, Victoria Land, and Wilkes Land, exhibit relatively weak changes, and some models even project decreases in precipitation in these areas (figure S4). To assess model uncertainty in future projections of Antarctic precipitation, we calculated the standard deviation across all models (figure 1(b)). Results show greater uncertainty in coastal regions than in the interior of Antarctica, with the spatial pattern of standard deviation closely resembling that of the precipitation amounts. To further evaluate the robustness of these projections, we computed the SNR, comparing the mean change to the standard deviation (figure 1(c)). Results show that uncertainty is highest in the coastal regions, where the SNR is less than 2, and is particularly pronounced in the western part of Antarctica, where the SNR is even below 1. In contrast, the interior of Antarctica exhibits relatively low uncertainty, with an SNR greater than 2. To provide context, we also analyzed projected changes in Antarctic surface temperature, which exhibit significantly less uncertainty, with SNR values exceeding 3 (figure S5).

Considering the differing climatology of precipitation across regions of Antarctica, we evaluated fractional changes to better investigate the regional features of Antarctic precipitation changes (figures 1(d)–(f)). The mean fractional increase in integrated Antarctic precipitation is 28%, with a wide range across models, from 9% to 49% (figure S6). Fractional precipitation changes are more pronounced in the interior of Antarctica, where increases exceed 40%, compared to the coastal regions. In contrast, the western part of West Antarctica, Victoria Land, and Wilkes Land exhibit relatively smaller changes, with increases of less than 20% (figure 1(d)). The standard deviation across models indicates greater uncertainty in the interior of Antarctica compared to the coastal regions (figure 1(e)). Despite differences



between the fractional and absolute changes, the SNR shows similar spatial patterns (figure 1(f)) to those of the absolute precipitation changes. Specifically, the coastal regions of Antarctica exhibit greater uncertainty, while the interior shows relatively lower uncertainty. The western part of West Antarctica, Victoria Land, and Wilkes Land remain areas of particularly high uncertainty, with significantly lower SNR values. These results also indicate that precipitation changes are associated with much greater uncertainty compared to surface temperature changes, which exhibit higher SNR values.

Given the differences in precipitation climatology among models, it is more meaningful to focus on fractional changes in precipitation. Accordingly, the remainder of this study emphasizes fractional changes in Antarctic precipitation and their associated model uncertainties. The SNR for the integrated fractional precipitation change is 2.5, compared to a SNR of 3.7 for the Antarctic mean surface temperature change. This highlights that model uncertainty in future projections of Antarctic precipitation is significantly larger than that of surface temperature and exhibits pronounced regional disparities. In the following sections, we explore the potential sources of model uncertainty in the future projections of both integrated and regional Antarctic precipitation.

3.2. Relationship to surface temperature and atmospheric circulation

To assess the relationship between changes in precipitation and the magnitude of global warming, we calculated the correlation between precipitation changes and the deviation of future global mean temperature projections from the multi-model mean (figures 2(a) and (b)). Results show that model uncertainty in global mean surface temperature accounts for 73% of the model uncertainty in integrated Antarctic precipitation changes (figure 2(a)). Given the large regional differences in precipitation changes (gray dot in figure 2(a)), it is necessary to evaluate the regional feature in the correlation between precipitation changes and global mean temperature changes. The regional correlation map shows a high correlation (~ 0.7) in the interior of East Antarctica, while the correlation is relatively low (< 0.5) in the coastal regions of West Antarctica and Wilkes Land (figure 2(b)). We also examined the correlation between precipitation changes and Antarctic mean surface temperature (figures 2(c) and (d)). The results reveal a stronger relationship, with model uncertainty in Antarctic mean surface temperature accounting for 95% of the model uncertainty in integrated Antarctic precipitation changes (figure 2(c)). The correlation map also reveals regional disparities, with much lower

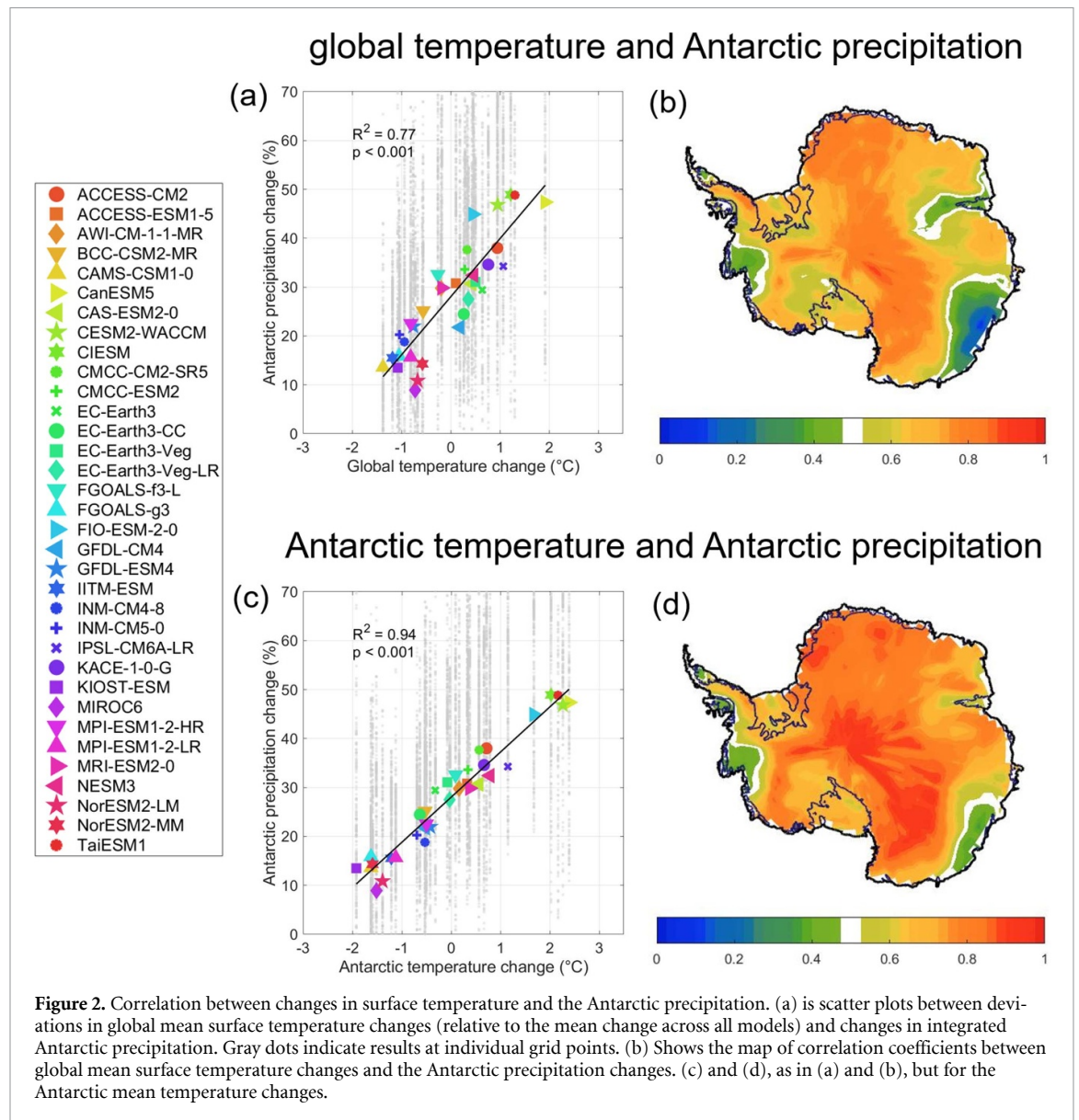


Figure 2. Correlation between changes in surface temperature and the Antarctic precipitation. (a) is scatter plots between deviations in global mean surface temperature changes (relative to the mean change across all models) and changes in integrated Antarctic precipitation. Gray dots indicate results at individual grid points. (b) Shows the map of correlation coefficients between global mean surface temperature changes and the Antarctic precipitation changes. (c) and (d), as in (a) and (b), but for the Antarctic mean temperature changes.

correlations in the coastal regions of West Antarctica and Wilkes Land compared to the interior of East Antarctica, despite an overall higher correlation coefficient relative to the results for global mean temperature (figure 2(d)). Nonetheless, the notably low correlations in West Antarctica and parts of East Antarctica highlight the potential contribution of atmospheric circulation effects. We also examined interannual temperature–precipitation covariability in ERA5 to ensure consistency with observations, and the results exhibit a similar spatial contrast (figure S7), supporting the interpretation that inland precipitation is more thermodynamically driven, while coastal precipitation is more dynamically controlled.

We further examined the relationship between precipitation changes and atmospheric circulation adjustments. First, to remove the influence of temperature on precipitation changes, we regressed the Antarctic precipitation changes onto Antarctic mean surface temperature changes and retained the regression residuals. The precipitation changes

exhibit more pronounced and diverse regional features among models after removing the temperature effect (figure S8). Second, we applied MCA to examine changes in Antarctic precipitation in conjunction with atmospheric circulation, using 500 hPa geopotential height in the mid-to-high latitudes of the Southern Hemisphere. The zonal mean of the 500 hPa geopotential height was removed to highlight regional variations. The first mode of the MCA results reveals an overall decrease in precipitation over West Antarctica and an increase over East Antarctica (figure 3(a)), accounting for 48% of the variance. In West Antarctica, the decreasing pattern is most pronounced in the coastal regions, extending from Mount Siple to the Antarctic Peninsula, except for a small area of increase in the western sector of West Antarctica. In East Antarctica, the increasing pattern is widespread, with the exception of a small decreasing area in the inland region of the Transantarctic Mountains. This west-east Antarctic precipitation pattern is closely associated with the PSA2 pattern

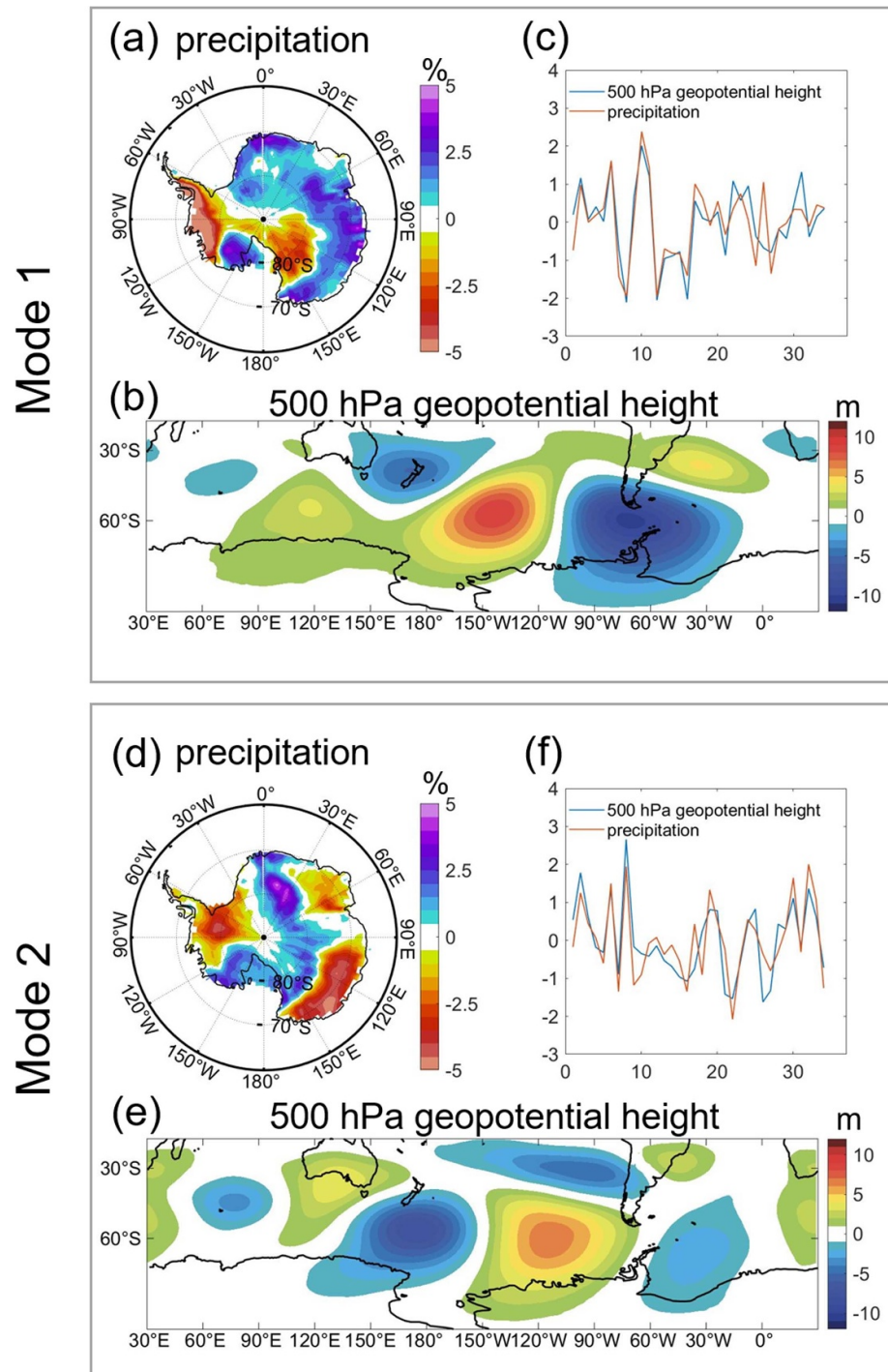
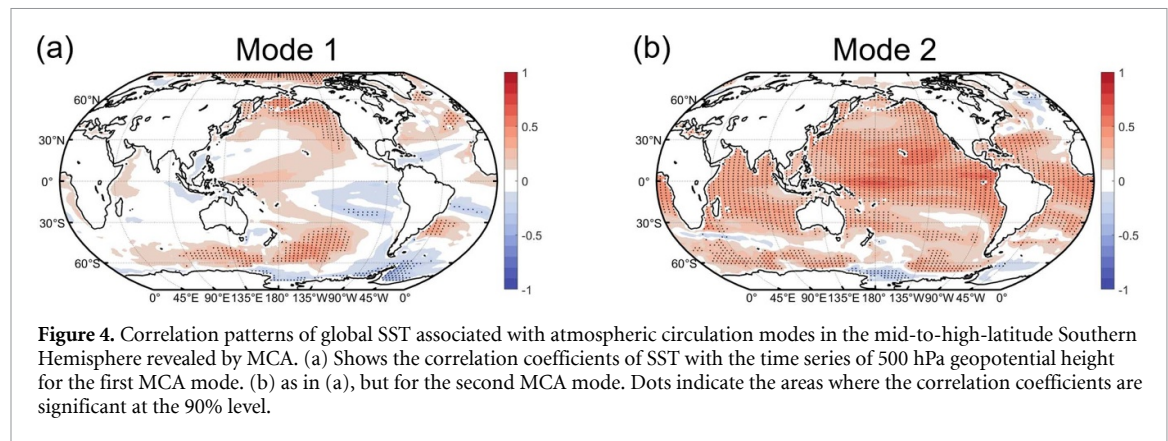


Figure 3. Linkages between changes in the Antarctic precipitation and atmospheric circulation in the mid-to-high-latitude Southern Hemisphere revealed by MCA. (a)–(c) show the first mode of MCA results: (a) precipitation changes, (b) 500 hPa geopotential height changes, and (c) normalized principal component (PC) time series. (d)–(f), as in (a)–(c), but for the second mode of MCA results.

in the Southern Hemisphere ($r = 0.85$, significant at the 99% level), characterized by low-pressure anomalies over the Drake Passage and New Zealand, and high-pressure anomalies over the Amundsen Sea and Dumont d'Urville Sea (figure 3(b)). To evaluate the similarity between the atmospheric circulation pattern and PSA2, we derived the PSA2 pattern from the CMIP6 multi-model composite in both historical (figure S9(b)) and SSP5-8.5 simulations (figure

S9(d)). The atmospheric circulation pattern identified in the first MCA mode shows a high spatial correlation with the PSA2 pattern derived from both the historical simulations ($r = 0.83$, significant at the 99% level) and the SSP5-8.5 simulations ($r = 0.85$, significant at the 99% level) (table S2). A similar correspondence is also found in ERA5 (figure S9(f)), where the first MCA mode exhibits a strong spatial correlation with the PSA2 pattern ($r = 0.84$),



providing observational support for the PSA2 circulation changes identified in the models.

The second mode of the MCA results reveals a more complex precipitation pattern across Antarctica (figure 3(d)), accounting for 25% of the variance. In West Antarctica, a dipole pattern emerges, with increased precipitation in the western part and decreased precipitation in the eastern part. Compared to the first mode, the decrease in West Antarctica is smaller in magnitude and more concentrated in the interior regions. In East Antarctica, the interior shows an increasing precipitation pattern, while the coastal regions, particularly in Wilkes Land, exhibit a pronounced decrease. This precipitation pattern is closely associated with the PSA1 pattern in the Southern Hemisphere ($r = 0.81$, significant at the 99% level), characterized by low-pressure anomalies over the Weddell Sea and Dumont d'Urville Sea, and high-pressure anomalies over the Amundsen–Bellingshausen Sea and south of Australia (figure 3(e)). This atmospheric pattern has a strong spatial correlation with the PSA1 pattern derived from both CMIP6 historical simulations ($r = 0.72$, significant at the 99% level, figure S9(a)) and SSP5-8.5 simulations ($r = 0.72$, significant at the 99% level, figure S9(c)). ERA5 also shows a closer correspondence between its second MCA mode and the PSA1 pattern ($r = 0.59$, figure S9(e)), further supporting the interpretation that PSA1 atmospheric variations contribute to the regional structure of precipitation changes.

In summary, differences in Antarctic surface temperature changes account for the model uncertainty in integrated precipitation changes over Antarctica, while differences in atmospheric circulation explain the regional features of model uncertainty in Antarctic precipitation. Specifically, the PSA2 pattern is associated with a contrasting precipitation pattern between West and East Antarctica, whereas the PSA1 pattern is linked to a dipole pattern in West Antarctica and an interior-versus-coastal pattern in East Antarctica. Previous studies suggest that the PSA1 and PSA2 patterns may originate from tropical SST anomalies [53–55]. In the following

sections, we explore the relationship between model uncertainty in tropical SST and Antarctic precipitation, as well as the associated atmospheric circulation patterns.

3.3. Influence of tropical teleconnections

We evaluated the future projections of tropical SST changes by the end of the 21st century (figure S10). The mean SST changes in tropical oceans are characterized by overall warming across the tropics, but with strong regional patterns resembling an El Niño-like pattern, where equatorial regions are warmer than the subtropical regions, particularly in the tropical Pacific (figure S10(a)). The model uncertainty in tropical SST also exhibits distinct regional patterns. The equatorial tropical Pacific shows the largest standard deviation among CMIP6 models compared to other tropical regions, while the subtropical regions of the tropical Pacific display relatively lower standard deviations (figure S10(b)). To further evaluate model uncertainty in tropical SST, we calculated the SNR (figure S10(c)). Results show the lowest values in the subtropical Pacific and the west-central equatorial Pacific, primarily due to small mean changes and high standard deviations, respectively. Overall, tropical SST projections show considerable model uncertainty with pronounced regional features, which may influence model uncertainty in Antarctic climate projections through atmospheric teleconnections.

To retrieve the contribution of SST changes to atmospheric circulation adjustments associated with Antarctic precipitation patterns, we regressed the global SST changes in CMIP6 onto the time series of the geopotential height for the two leading modes (figure 4). Results show that the PSA2 pattern in the first mode is associated with a west-warming-east-cooling pattern of tropical SST (figure 4(a)). The significant areas are located where the SNR of SST is lowest in the tropics, suggesting that model uncertainty in tropical SST may contribute to the PSA2 atmospheric circulation pattern and the associated contrasting precipitation pattern between West and East Antarctica. Nonetheless, the relatively low correlation coefficients between tropical SST and the first

mode also suggest that internal atmospheric variability plays a role. On the other hand, the PSA1 pattern in the second mode is linked to a widespread warming of tropical SST, with pronounced warming in the western-central equatorial Pacific (figure 4(b)), where the SNR of SST is relatively low. This result indicates that model uncertainty in the El Niño-like pattern in the tropical Pacific, along with the warming patterns in the tropical Indian and Atlantic Oceans, may contribute to the PSA1 atmospheric circulation pattern, as well as the dipole precipitation pattern in West Antarctica and the interior-coastal precipitation pattern in East Antarctica.

4. Conclusions

In this study, we analyzed the model uncertainty in future projections of Antarctic precipitation under the high-emission scenario (SSP5-8.5) using CMIP6 models. Our results show that the Antarctic precipitation is projected to increase significantly by the end of the 21st century, with significant regional disparities. Importantly, precipitation projection uncertainties are substantially larger than those for surface temperature, particularly in coastal regions and the western parts of West Antarctica. Our analysis shows that much of the uncertainty in integrated Antarctic precipitation projections can be attributed to model variability in global and Antarctic surface temperatures. However, regional differences in precipitation changes are strongly influenced by variability in atmospheric circulation patterns, especially those linked to the PSA1 and PSA2 modes. Moreover, model uncertainty in tropical SST contributes to adjustments in atmospheric circulation, further impacting Antarctic precipitation projections.

These findings highlight the critical challenges in accurately projecting Antarctic precipitation and its implications for future sea-level rise. The pronounced uncertainty in coastal precipitation projections underscores the urgent need to improve the representation of polar processes in global climate models. Given that regional Antarctic precipitation is a key climate forcing in ice sheet models, further research is essential to better constrain the impacts of these uncertainties on ice sheet dynamics and long-term sea-level projections. Additionally, the connection between Antarctic precipitation and tropical oceans, mediated through atmospheric circulation patterns, suggests that biases or uncertainties in tropical SSTs could have cascading effects on regional Antarctic climate projections—effects that may be obscured by large-scale changes. Addressing these uncertainties, particularly those related to atmospheric circulation variability and the influence of tropical SSTs on Antarctic climate, should remain a key priority for future research.

Data availability statement

The data from CMIP6 models are available at the Earth System Grid Federation (ESGF) (<https://aims2.llnl.gov/search/cmip6/>). The processed data were regridded to a $1^\circ \times 1^\circ$ grid using Climate Data Operators (CDO), which is freely available at <https://code.mpimet.mpg.de/projects/cdo>.

Supporting figures and tables available at <https://doi.org/10.1088/1748-9326/ae2ca9/data1>.

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Conceptualization (equal), Data curation (lead), Formal analysis (lead), Investigation (lead), Methodology (lead), Resources (lead), Software (lead), Validation (lead), Visualization (lead), Writing – original draft (lead), Writing – review & editing (lead)

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